Phase 2 of project2: Innovation of CRL in locomotion task and TD3 in manipulation task

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April 2026

1 Introduction

We focus on enhancing contrastive reinforcement learning (CRL [4]) in the locomotion task and twin delayed deep deterministic policy gradient (TD3 [5]) in the manipulation task on JaxGCRL [2]. For CRL, we first strengthen the baseline with a twin-Q critic [6] and then further improve it with a cross-episode HER [1] sampling mechanism. For TD3, we incorporate a TQC-inspired quantile critic with quantile Huber regression, while setting the truncation drop to zero in practice.

1.1 Manipulation vs Locomotion Environments

The JaxGCRL benchmark evaluates agents across two distinct categories of control: locomotion and manipulation. While both utilize goal-conditioned reinforcement learning, they present different challenges to the agent's policy. Locomotion tasks in this project involve a quadruped "Ant" agent navigating through various spaces to reach a target position. These environments, such as Ant U-Maze and Ant Big Maze, test an agent's ability to coordinate multiple joints for consistent movement. The difficulty in these tasks primarily arises from physical barriers like maze walls which require the agent to learn paths where walking straight to a goal may not be the most feasible.

Manipulation tasks, specifically the Arm Push Hard environment, involve a robotic arm that has to push/move objects to goal locations. Unlike locomotion, manipulation requires more precise, delicate control and an understanding of object dynamics. These tasks are characterized by higherdimensional state spaces and a "contact-rich" nature, where small errors in the arm's trajectory can lead to significant failures in object placement.

1.2 CRL Background

Contrastive Reinforcement Learning (CRL) is a framework tailored for goal-conditioned environments, designed to operate efficiently under sparse, binary rewards. Rather than minimizing standard temporal difference (TD) errors, CRL reframes value estimation as a self-supervised representation learning problem. It utilizes a contrastive objective—typically the InfoNCE loss—to measure the temporal and spatial compatibility between a state-action pair and a future goal. Within a learned latent space, the algorithm maximizes similarity between state-action embeddings and the goals they successfully reach, while explicitly pushing them apart from uncorrelated goals in the batch. This contrastive matching provides a dense signal that drives efficient exploration.

1.3 TD3 Background

Twin Delayed Deep Deterministic Policy Gradient (TD3) is an actor-critic algorithm specifically designed for continuous control spaces. TD3 implements Clipped Double Q-Learning, utilizing two independent critics and bounding the target Q-value with the minimum of their estimates. Second, it utilizes Delayed Policy Updates, optimizing the actor network and updating target networks less frequently than the critics to ensure the policy is trained against a stabilized value function. Finally, TD3 applies Target Policy Smoothing by injecting clipped Gaussian noise into the target actions, which prevents it from exploiting sharp peaks in the Q-function.

2 Contributions.

- **Twin-Q stabilization for contrastive reinforcement learning.**

  We first extend the original CRL framework with a twin-Q critic design. Two critics are trained in parallel, and the actor is optimized using the minimum of their Q estimates. This reduces overestimation and provides a more stable value-learning signal.

- **A nearest-neighbor cross-episode goal relabeling mechanism for CRL.**

Building on the twin-Q CRL variant, we further introduce a nearest-neighbor cross-episode goal relabeling mechanism. For each sampled transition, we first identify the most similar state in the current mini-batch using Euclidean distance. Then, with a fixed probability, we replace the current goal with the neighbor's relabeled goal, and simultaneously replace the associated future state and future action with those from the same neighbor transition. This alleviates the penalty imposed by contrastive learning on those future scenarios that could be achieved but were not sampled in the replay buffer.

- **A quantile-regression-based distributional critic for TD3.**

  For the manipulation task, we replace the scalar TD3 critic with a quantile critic that outputs multiple return quantiles for each state-action pair, so that the Q-value is estimated from a set of quantile predictions rather than a single scalar. The critic is trained with quantile Huber regression, while the overall TD3 training pipeline, including delayed policy updates and target smoothing, is preserved. In our experiments, we retain the full target quantile set rather than applying truncation like TQC [3], so the main effect of this modification is to enrich value modeling through distributional supervision instead of scalar Q regression.

3 Methodology.

3.1 Twin-Q in CRL

Twin-Q contrastive critic in CRL. For each training sample, let s denote the state, a the action, and g the goal relabeled by HER. Instead of using a single contrastive critic, we use two independent critics:

$$\phi_1(s,a) \in \mathbb{R}^d, \qquad \psi_1(g) \in \mathbb{R}^d,$$

$$\phi_2(s,a) \in \mathbb{R}^d, \qquad \psi_2(g) \in \mathbb{R}^d.$$

Each critic defines an energy-based goal-conditioned value score

$$Q_k(s,a,g) = \mathcal{E}\left(\phi_k(s,a), \psi_k(g)\right), \qquad k \in \{1,2\}.$$

In our implementation, the default energy function is the negative Euclidean distance:

$$\mathcal{E}(x,y) = -\sqrt{\|x-y\|_2^2 + \varepsilon}.$$

For a mini-batch $\{(s_i, a_i, g_i)\}_{i=1}^{B}$, each critic forms a logits matrix $\ell^{(k)} \in \mathbb{R}^{B\times B}$, where the element of the matrix is

$$\ell_{ij}^{(k)} = \mathcal{E}(\phi_k(s_i, a_i), \psi_k(g_j)), \qquad k \in \{1, 2\}.$$

Using the forward InfoNCE objective, the contrastive loss for critic k is

$$\mathcal{L}_{\text{InfoNCE}}^{(k)} = -\frac{1}{B}\sum_{i=1}^{B}\left(\ell_{ii}^{(k)} - \log\sum_{j=1}^{B}\exp\left(\ell_{ij}^{(k)}\right)\right).$$

Define the row-wise log-sum-exp terms

$$r_i^{(1)} = \log\sum_{j=1}^{B}\exp\left(\ell_{ij}^{(1)}\right), \qquad r_i^{(2)} = \log\sum_{j=1}^{B}\exp\left(\ell_{ij}^{(2)}\right).$$

Then the critic objective used in the code is

$$\mathcal{L}_{\text{critic}} = \frac{1}{2}\left(\mathcal{L}_{\text{InfoNCE}}^{(1)} + \mathcal{L}_{\text{InfoNCE}}^{(2)}\right) + \lambda_{\text{lse}} \cdot \frac{1}{B}\sum_{i=1}^{B}\left(\frac{r_i^{(1)} + r_i^{(2)}}{2}\right)^2.$$

Twin-Q actor update. Given a goal-conditioned policy $\pi_\theta(a \mid s, g)$, we sample

$$u \sim \mathcal{N}\left(\mu_\theta(s, g),\, \sigma_\theta^2(s, g)\right), \qquad a = \tanh(u).$$

Its log-density is

$$\log \pi_\theta(a \mid s, g) = \sum_m \left[\log \mathcal{N}\left(u_m; \mu_m, \sigma_m^2\right) - \log\left(1 - a_m^2 + \varepsilon\right)\right].$$

The second term is the Jacobian correction induced by the tanh transformation, which accounts for the change in probability density when mapping the Gaussian sample u to the bounded action $a = \tanh(u)$. The two critics produce

$$Q_1(s, a, g) = \mathcal{E}(\phi_1(s, a), \psi_1(g)), \qquad Q_2(s, a, g) = \mathcal{E}(\phi_2(s, a), \psi_2(g)),$$

and the actor uses the more conservative estimate

$$Q_{\min}(s, a, g) = \min\{Q_1(s, a, g), Q_2(s, a, g)\}.$$

The policy objective is

$$\mathcal{L}_{\text{actor}} = \frac{1}{B}\sum_{i=1}^{B}\left[\alpha \log \pi_\theta(a_i \mid s_i, g_i) - Q_{\min}(s_i, a_i, g_i)\right].$$

This objective trains the policy to maximize the conservative value estimate given by the twin critics, while preserving exploration through entropy regularization.
Entropy coefficient update. Let $\bar{\mathcal{H}}$ denote the target entropy. The temperature parameter is updated by

$$\mathcal{L}_\alpha = \alpha \cdot \frac{1}{B} \sum_{i=1}^{B} \left(-\log \pi_\theta(a_i \mid s_i, g_i) - \bar{\mathcal{H}} \right), \qquad \alpha = \exp(\log \alpha).$$

3.2 Nearest-neighbor cross-episode goal relabeling in CRL

After constructing the replay batch, we augment the relabeling procedure rather than the contrastive objective itself. For each transition i, let s_i denote its current state. We first find the nearest state in the current mini-batch:

$$j^*(i) = \arg\min_{j \neq i} \|s_i - s_j\|_2^2.$$

Each sampled transition already contains a relabeled goal g_i, a future state s_i^+, and a future action a_i^+. We then draw a Bernoulli variable

$$z_i \sim \mathrm{Bernoulli}(p),$$

Here, p denotes the probability of applying cross-episode relabeling. Then we perform the following replacement:

$$(\tilde{g}_i, \tilde{s}_i^+, \tilde{a}_i^+) = \begin{cases} (g_{j^*(i)},\, s_{j^*(i)}^+,\, a_{j^*(i)}^+), & z_i = 1, \\ (g_i,\, s_i^+,\, a_i^+), & z_i = 0. \end{cases}$$

The relabeled observation used by the critic becomes

$$\tilde{o}_i = [\, s_i, \tilde{g}_i \,].$$

At the same time, the stored future supervision is updated to

$$(\tilde{s}_i^+, \tilde{a}_i^+),$$

so that the actor and critic remain conditioned on consistent relabeled future information.

This mechanism does not change the CRL loss itself. Instead, it modifies how replayed goals and future supervision are constructed, reduce negative effects of future assignments with big gap for nearby states under contrastive training. In standard contrastive pairing, similar states may be repeatedly associated with different future targets, which can separate behaviorally compatible trajectories. By allowing nearby states to share future supervision across episodes, our cross-episode relabeling introduces a local smoothing effect on goal assignment and yields more stable training signals.

3.3 Quantile-regression-based critic in TD3.

For the manipulation task, we preserve the original TD3 training structure and replace the scalar critic with a quantile critic. For each state-action pair (s, a), critic k outputs a set of N quantiles

$$Z_k(s,a) = \{q_{k,1}(s,a), \dots, q_{k,N}(s,a)\}, \qquad k = 1, \dots, n_{\text{critics}},$$

where N is the number of quantiles predicted by each critic, and n_{critics} is the number of parallel quantile critics. Thus, instead of estimating a single scalar Q-value, the critic represents the return distribution through multiple quantile estimates.

The target action is computed using TD3-style target policy smoothing:

$$a' = \text{clip}\big(\pi_{\bar{\phi}}(s') + \epsilon,\, -a_{\max},\, a_{\max}\big)\,, \qquad \epsilon \sim \text{clip}\big(\mathcal{N}(0,\sigma^2),\, -c,\, c\big)\,.$$

The target critic then outputs target quantiles at (s', a'), and the Bellman targets are written as

$$y_n = r + \gamma \bar{y}_n, \qquad n = 1, \dots, N',$$

where $\{\bar{y}_n\}_{n=1}^{N'}$ denotes the target quantile set.

The critic is trained with a quantile regression (QR) loss as

$$\mathcal{L}_{\text{critic}} = \frac{1}{n_{\text{critics}}} \sum_{k=1}^{n_{\text{critics}}} \sum_{m=1}^{N} \frac{1}{N'} \sum_{n=1}^{N'} \rho^{\kappa}_{\tau_m}\left(y_n - q_{k,m}(s,a)\right),$$

where the quantile levels are

$$\tau_m = \frac{2m-1}{2N}, \qquad m = 1, \dots, N,$$

and the quantile Huber loss [3] is defined as

$$\rho^{\kappa}_{\tau}(u) = |\tau - \mathbf{1}(u < 0)|\,\frac{\mathcal{H}_\kappa(u)}{\kappa},$$

with

$$\mathcal{H}_\kappa(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \kappa, \\ \kappa\left(|u| - \frac{1}{2}\kappa\right), & |u| > \kappa. \end{cases}$$

In this objective, each predicted quantile $q_{k,m}(s,a)$ is compared against the target quantiles $\{y_n\}_{n=1}^{N'}$, so the critic learns the return distribution through full pairwise quantile matching rather than scalar value regression. Therefore, the main modification is the introduction of quantile-regression-based distributional value estimation into the original TD3 framework.

4 Experiment

In this section, we evaluate the performance of our proposed innovations against the baselines established in Phase 1. For locomotion tasks, we introduce Twin-Q stabilization and Cross-Episode HER, while for manipulation tasks, we replace the original TD3 critic loss with QR loss. Across the evaluated environments, these modifications generally improve learning efficiency and final success rates compared with the corresponding baselines.

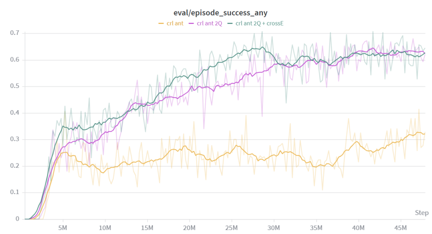
4.1 Hyperparameter design.

For the CRL experiments, we set the cross-episode relabeling probability to $p = 0.001$ for Ant and Ant Soccer, and to $p = 0.1$ for Ant U-Maze, Ant Push, and Ant Big Maze. We use a larger p in environments with stronger structural constraints or more repetitive local state patterns, where nearby states are more likely to share compatible future supervision. For the QR-loss-based manipulation experiments, we use $n_{\text{critics}} = 5$ critics and $N = 25$ quantiles per critic, which provides a richer distributional value estimate while keeping the computational cost manageable.

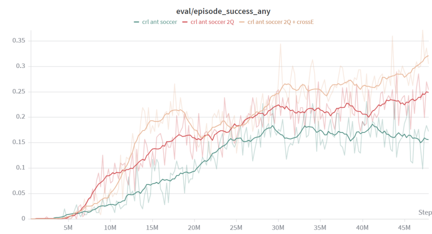
4.2 Result Analysis

As shown in Fig. 1(a)–(e) and Table 1, the proposed modifications consistently improve CRL-based locomotion performance across all five environments, although the degree of improvement depends on task structure. In the standard Ant environment, the best success rate increases sharply from 0.327 for the baseline to 0.645 with Twin-Q, while Cross-Episode HER only brings a very small additional increase to 0.648. This suggests that in relatively open environments, critic stabilization is the main source of improvement, whereas cross-episode relabeling provides limited extra benefit.

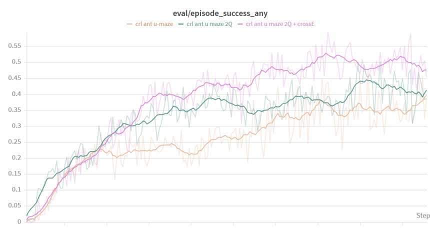
In contrast, environments with stronger structural constraints show clearer gains from combining both components. In Ant Soccer, performance rises from 0.185 to 0.250 with Twin-Q and further to 0.321 with Cross-Episode HER. In Ant U-Maze, the success rate improves from 0.398 to 0.445 and then to 0.529.



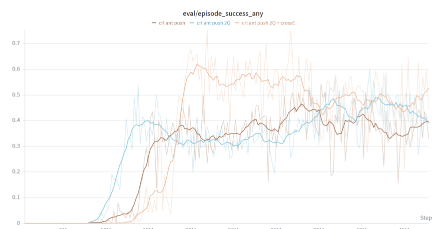
(a) crl ant



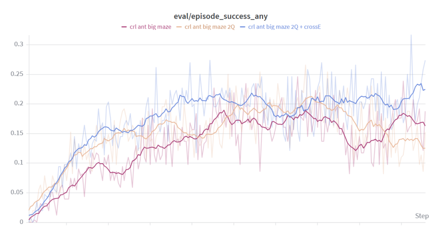
(b) crl ant soccer



(c) crl ant u maze



(d) crl ant pusher



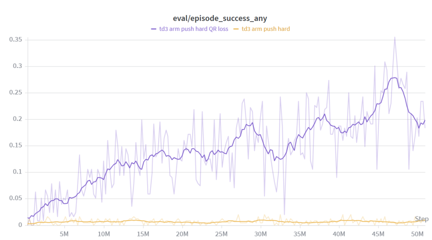
(e) crl ant big maze

Figure 1: Performance comparison of CRL-based methods across five locomotion environments using best success rate.

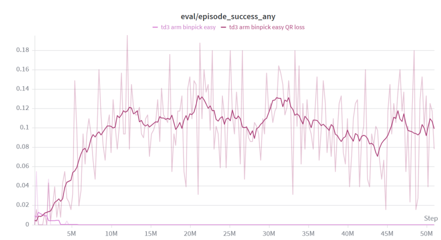
Table 1: Locomotion performance comparison using best success rate.

Environment	Baseline (CRL)	+ Twin-Q	+ Twin-Q + Cross-Episode HER
Ant	0.327	0.645	0.648
Ant Soccer	0.185	0.250	0.321
Ant U-Maze	0.398	0.445	0.529
Ant Push	0.462	0.490	0.620
Ant Big Maze	0.189	0.212	0.234

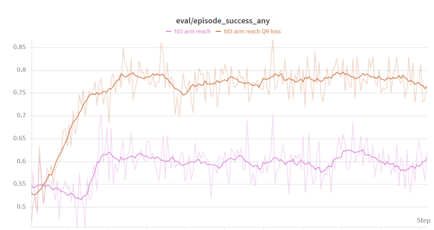
A similar pattern is also observed in Ant Push, where the baseline score of 0.462 increases to 0.490 with Twin-Q and then to 0.620 after adding Cross-Episode HER. Even in Ant Big Maze, where exploration is more difficult and the overall success rate remains lower, the same trend still appears, with performance improving from 0.189 to 0.212 and then to 0.234. Overall, these results indicate that Twin-Q provides a stable and reliable improvement across locomotion tasks, while Cross-Episode HER is especially effective in environments where agents repeatedly encounter similar geometric bottlenecks, interaction patterns, or state-goal relationships.



(a) td3 arm push hard



(b) td3 arm binpick easy



(c) td3 arm reach

Figure 2: Performance comparison of TD3-based methods across three manipulation environments using best success rate.

As shown in Fig. 2(a)–(c) and Table 2, the modified TD3-based method

Table 2: Manipulation performance comparison using best success rate.

Environment	Baseline (TD3)	+ QR Loss
Arm Push Hard	0.010	0.279
Arm Binpick Easy	0.013	0.133
Arm Reach	0.626	0.796

also achieves consistent improvements across all three manipulation environments. In Arm Push Hard, the best success rate increases substantially from 0.010 to 0.279, showing that the new method is much more effective than the baseline in this difficult sparse-reward setting. In Arm Binpick Easy, performance improves from 0.013 to 0.133, which again suggests a clear benefit in tasks that require more precise exploration and credit assignment. Even in the easier Arm Reach task, where the baseline already performs reasonably well at 0.626, the modified method still raises the success rate to 0.796. Compared with the CRL results, the TD3 improvements appear more uniformly distributed across the three tasks, suggesting that the proposed modification improves optimization quality and robustness not only in hard manipulation problems but also in relatively simple goal-reaching settings. Taken together, the results in Fig. 2(a)–(c) and Table 2 show that the modified TD3 framework consistently enhances final task performance and provides a stronger baseline for manipulation environments.

5 Conclusion

In this work, we improved CRL for locomotion and TD3 for manipulation. For locomotion, Twin-Q provided a more stable value-learning signal, while Cross-Episode HER further improved performance in environments with stronger structural constraints. For manipulation, replacing the scalar TD3 critic with a quantile-regression-based critic produced consistent gains across all evaluated tasks. Overall, the results show that the proposed modifications improve both training stability and final task performance. The idea of Cross-Episode was inspired by *Teamfight Tactics*: when the opening draw does not support the originally planned trait, players often sell units to a better-supported composition. At this moment, player need to align the current state with the most likely future trajectory with highest rewards. Similarly, Cross-Episode allows a transition to reuse future supervision from a nearby compatible transition. If one path seems unfeasible, one will choose future trajectory of another state that is the closest in distance. Without practicing this technique, you will keep losing when you are not fortunate enough, so it is essential to train the transition process.

We also explored many unsuccessful variants, including BCE-style CRL losses, stronger direct goal supervision in CRL, learnable or distance-based transition probabilities, TQC-style designs inside CRL's twin Q, original TQC with quantile dropping, and QR loss combined with Distributional NCE. These negative results were also valuable, because they helped identify the constraint of the contrastive learning and manipulation task. As a result, the true experimental workload of this project was roughly three times larger than what the final report alone suggests.

6 References

References

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